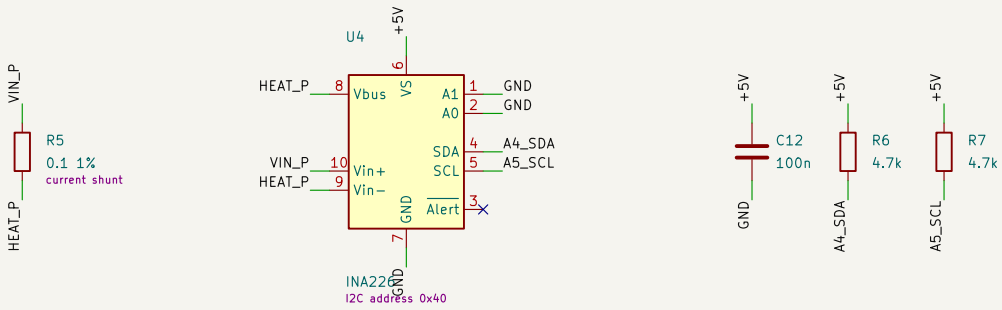
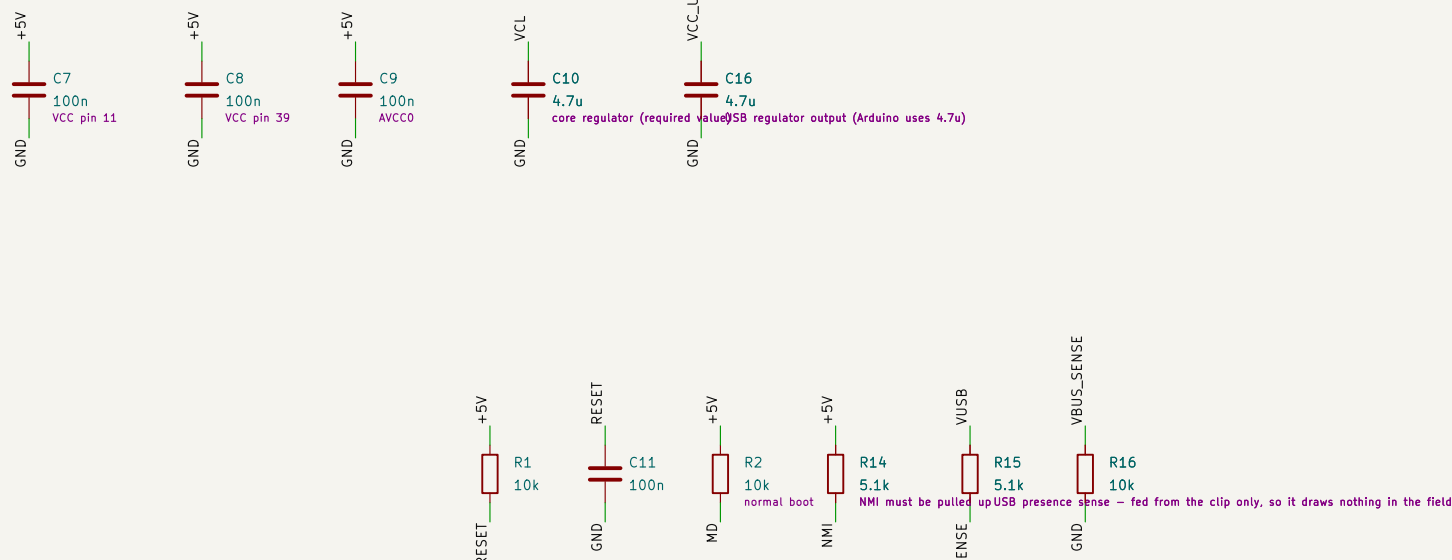
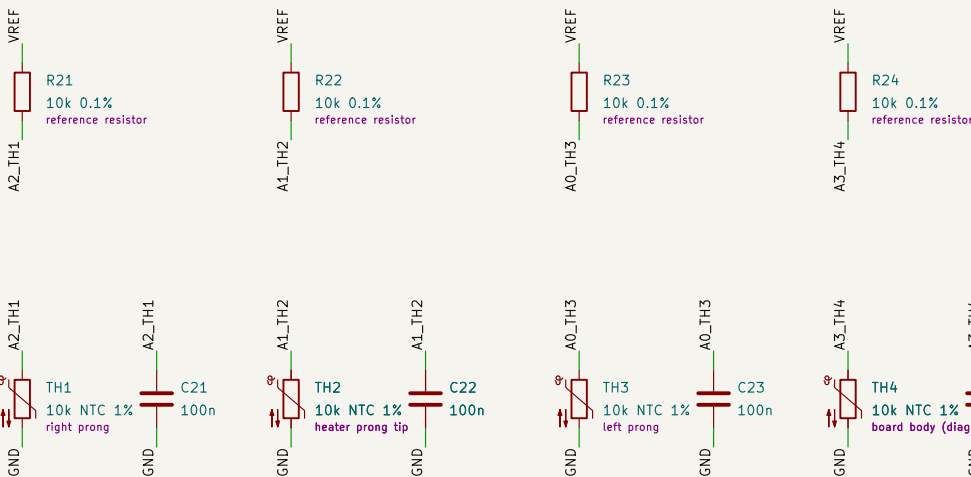


SDI-12 DATA – meets the SDI-12 electrical numbers: R3 sets the 1–2 k transmit resistance (and limits fault current into pin D2), R4 the 160–360 k idle resistance, C4 holds the edge rate under 1.5 V/us. D4 clamps cable transients.

HEATER_DRIVE + POWER_MONITOR PWM-capable D9/P303 drives Q1. Required
firmware: 100 Hz, nominat 2 V average; 8–15 s; integrate stable ON power times
actual ON time. Force D9 LOW at shutdown; stopping a timer is insufficient. R10
pulls an undriven gate LOW. Keep all three INA226 inputs on their dedicated
Kelvin connections. Firmware is not included.

[illegible]

THERMISTORS – pipamp (same on flat-nano and atel); A2 = TH1 right needle, A1 = TH2 heater-needle tip, A0 = TH3 left needle, A3 = TH4 board body (left/right seen on the top face, cable end up, needles down). Required firmware: 14-bit ADC, external VREF (AR_EXTERNAL), 64 readings averaged; actual noise and accuracy require measurement. The dividers and ADC reference share VREF. D4 powers the divider and ADC; D0 powers the heater, so that the VREF is 1.1V. D5 powers the self-heating and the reading stays atmospheric. Sequence: D4 HIGH, wait 10 ms, read, D4 LOW, R_ntc = R_ref * code / (full scale - code). NTC = Murata NCP15X1H03F03RC (10k, 1%); R_ref = 10k 0.1% 25 ppm.



J2



A diagram of a 5-pin connector labeled J2. The pins are numbered 1 to 5 from top to bottom. Pin 1 is labeled VUSB, pin 2 is labeled USB_DM, pin 3 is labeled USB_DP, pin 4 is labeled GND, and pin 5 is labeled MD. The connector is shown as a yellow rectangular block with five pins extending from it.

programming clip holes
1 = SV, 2 = D-, 3 = D+, 4 = GND, 5 = BOOT

Generated by gen_schematic.py - edit the script, not this file

K-State Soil Water Processes Lab

Sheet: /

File: hp_sensor.kicad_sch

Title: Compact heat-pulse soil sensor, SDI-12 (HP-SDI12-NANO)

Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 1/1

